

JACOB BAKER-SPITZER

Durham, NC

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Education

Duke University, Pratt School of Engineering

Aug 2026 – May 2030 (Expected)

Bachelor of Science in Engineering, Electrical & Computer Engineering

Durham, NC

Relevant Coursework

- Matrices and Vectors
- Computing for Engineers
- Engineering Design
- Economic Principles

Experience

Research Assistant

Jun 2026 – Jul 2026

Economics Research Project

New York, NY

- Built a building dataset with 1,000+ entries spanning about 100 years of NYC construction, later used in an economics research paper
- Compiled structural data (floors, elevators, fireproofing) from historical fire records in Google Sheets over 4 weeks
- Verified lot labels against a single source of truth to maintain database accuracy

Engineering Shadow

Aug 2025 – Aug 2025

LiRo Group

New York, NY

- Documented the roof support systems and structural condition of 3 buildings with photos, notes, and field measurements

New Member Education Cohort

Sep 2026 – Present

Catalyst

Durham, NC

- Selected as 1 of 20 from roughly 200 applicants through a multi-round recruitment process

Black Belt (First Degree) & Assistant Instructor

Aug 2013 – Present

Riverdale Kenshikai Karate

Bronx, NY

- Earned a black belt in over 13 years of training and 600+ classes
- Co-taught 50+ classes of 10–20 students alongside other black belts by leading drills, correcting form, and preparing students for belt promotions

Certified Youth Soccer Referee

Sep 2023 – Jun 2026

Westchester Youth Soccer League

Westchester, NY

- Officiated 180+ U9–U12 matches over 3 years, making real-time calls and managing players, coaches, and parents
- Advanced from assistant referee to center referee

Incoming Engineering Trainer, Light/Energy Team

Jan 2027 – May 2027

Ignite

Durham, NC

Projects

Adaptive Fishing Game | *Assistive Technology, Design Process*

Aug 2026 – Present

- Defined requirements for a 2-switch fishing game for a child with limited mobility by interviewing the client and his care team with a 5-person design team

Hands-On Engineering Builds | *Soldering, 3D Printing*

Aug 2026 – Sep 2026

- Soldered an LED switching circuit for a lighted sign and 3D-printed a custom mug with the Duke logo

Technical Skills

Languages: Python

Developer Tools: VS Code, Git/GitHub, Arduino IDE, Onshape (CAD), Google Sheets

Electronics & Prototyping: Soldering, Circuit Prototyping, 3D Printing